

Research Notes on Variational autoencoder

Variational autoencoder

$$\mathcal{L} = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{\text{KL}}(q_\phi(z|x) \| p(z))$$

>> Section 1

Formulation From the point of view of probabilistic modeling, one wants to maximize the likelihood of the data x by their chosen parameterized probability distribution $p_\theta(x) = p(x|\theta)$. This distribution is usually chosen to be a Gaussian $N(x|\mu, \sigma)$ which is parameterized by μ and σ respectively, and as a member of the exponential family it is easy to work with as a noise distribution. Simple distributions are easy enough to maximize, however distributions where a prior is assumed over the latents z results in intractable integrals. Let us find $p_\theta(x)$ via marginalizing over z .

>> Section 2

$$L_\theta, \phi = \mathbb{E}_{x \sim \text{P real}} [\|x - D_\theta(E_\phi(x))\|^2] + d(\mu(dz), E_\phi(\text{P real}))^2 \\ = \mathbb{E}_{x \sim \text{P real}} [\|x - D_\theta(E_\phi(x))\|^2] + d(\mu(dz), E_\phi(\text{P real}))^2$$

>> Section 3

The statistical distance d requires special properties, for instance it has to be possess a formula as expectation because the loss function will need to be optimized by stochastic optimization algorithms. Several distances can be chosen and this gave rise to several flavors of VAEs:

>> Section 4

Unfortunately, the computation of $p_\theta(z|x)$ is expensive and in most cases intractable. To speed up the calculus to make it feasible, it is necessary to introduce a further function to approximate the posterior distribution as

>> Section 5

This can be reparametrized by letting $\epsilon \sim N(0, I)$ be a "standard random number generator", and construct z as $z = \mu_\phi(x) + L_\phi(x) \epsilon$. Here, $L_\phi(x)$ is obtained by the Cholesky decomposition: $\Sigma_\phi(x) = L_\phi(x)$

$\Sigma_\phi(x) = L_\phi(x) L_\phi(x)^T$. Then we have $\nabla_\phi \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(x, z) q_\phi(z|x)] = \mathbb{E} [\nabla_\phi \ln p_\theta(x, \mu_\phi(x) + L_\phi(x) \epsilon) q_\phi(\mu_\phi(x) + L_\phi(x) \epsilon | x)]$ and so we obtained an unbiased estimator of the gradient, allowing stochastic gradient descent. Since we reparametrized z , we need to find $q_\phi(z|x)$. Let q_0 be the probability density function for ϵ , then $\ln q_\phi(z|x) = \ln q_0(\epsilon) - \ln |\det(\partial \epsilon / \partial z)|$ where $\partial \epsilon / \partial z$ is the Jacobian matrix of z with respect to ϵ . Since $z = \mu_\phi(x) + L_\phi(x) \epsilon$, this is $\ln q_\phi(z|x) = -\frac{1}{2} \epsilon^T \epsilon - \ln |\det L_\phi(x)| - \frac{n}{2} \ln (2\pi)$

>> Section 6

In machine learning, a variational autoencoder (VAE) is an artificial neural network architecture introduced by Diederik P. Kingma and Max Welling in 2013. It is part of the families of probabilistic graphical models and variational Bayesian methods. In addition to being seen as an autoencoder neural network architecture, variational autoencoders can also be studied within the mathematical formulation of variational Bayesian methods, connecting a neural encoder network to its decoder through a probabilistic latent space (for example, as a multivariate Gaussian distribution) that corresponds to the parameters of a variational distribution. Thus, the encoder maps each point (such as an image) from a large complex dataset into a distribution within the latent space, rather than to a single point in that space. The decoder has the opposite function, which is to map from the latent space to the input space, again according to a distribution (although in practice, noise is rarely added during the decoding stage). By mapping a point to a distribution instead of a single point, the network can avoid overfitting the training data. Both networks are typically trained together with the usage of the reparameterization trick, although the variance of the noise model can be learned separately. Although this type of model was initially designed for unsupervised learning, its effectiveness has been proven for semi-supervised learning and supervised learning.

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>> Section 7

$$\begin{aligned} D_{KL}(q_\phi(z|x) \parallel p_\theta(z|x)) &= \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln \frac{q_\phi(z|x)}{p_\theta(z|x)}] = \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln q_\phi(z|x) - \ln p_\theta(z|x)] \\ &= \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln q_\phi(z|x)] - \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(z|x)] \\ &= \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln q_\phi(z|x)] - \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(x, z)] \\ &= \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln q_\phi(z|x)] - \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(x) + \ln p_\theta(z|x)] \\ &= \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln q_\phi(z|x)] - \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(x)] - \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(z|x)] \\ &= \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln q_\phi(z|x)] - \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(x)] - \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(z|x)] \end{aligned}$$